

به نام خالق زیبایی ها



درس رمزنگاری

گزارش سوال ۶ تکلیف سری اول

نگارنده : مهیار عنصری

تاریخ نگارش : اسفند ماه ۱۳۹۹

1. Encrypt your full name using the Caesar cipher with key = 'M' (to do so select Crypt/Decrypt > Symmetric (classic) > Caesar). how many letters is the alphabet shifted by?

Key Entry: Caesar / ROT-13

Description
Here you can enter the key for the Caesar cipher.
Caesar is a mono-alphabetic substitution, where the characters of the cleartext alphabet are mapped to the ciphertext alphabet by shifting. This shifting value is the key. You can enter the key as a number or as a single character of the alphabet.
Rot-13 is a special variant, where the key has the fixed value of half the length of the cleartext alphabet. This variant is only selectable if the length of the alphabet is an even number.

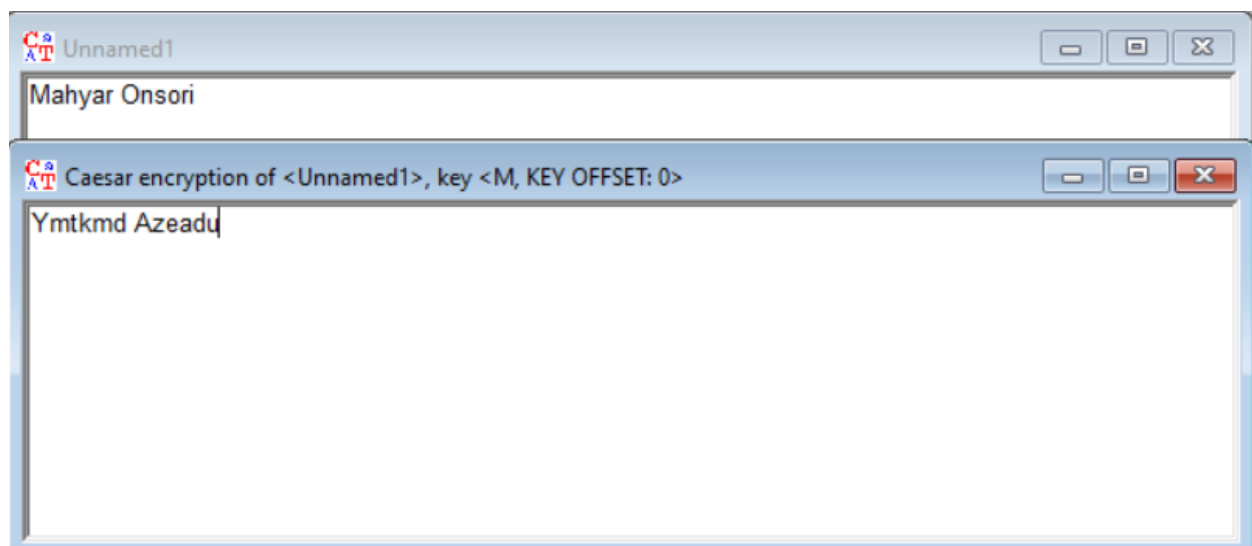
Select variant
☒ Caesar
☐ Rot-13

Options to interpret the alphabet characters
☒ Value of the first alphabet character = 0 (e.g. "A"=0)
☐ Value of the first alphabet character = 1 (e.g. "A"=1)

Key entry as
☒ Alphabet character
☐ Number value

Properties of the chosen encryption
Shift of 12
Mapping of the alphabet (26 characters)
from: ABCDEFGHIJKLMNOPQRSTUVWXYZ
to: MNOPQRSTUVWXYZABCDEFGHIJKL

Encrypt Decrypt Text options Cancel



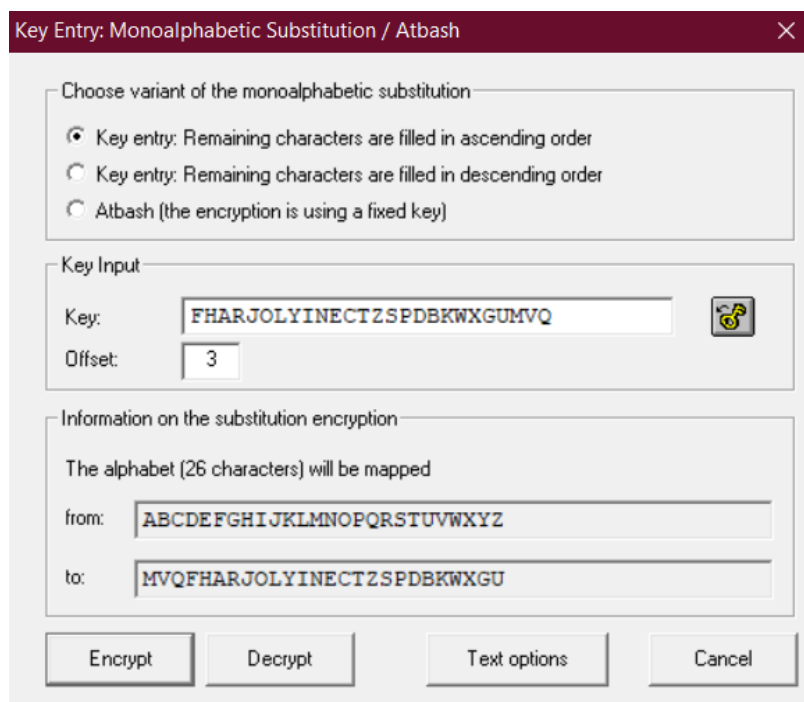
در این سوال چون M حرف سیزدهم و A حرف اول الفبای انگلیسی می باشد پس همه حروف به اندازه ۱۲ تا شیفت پیدا می کنند و به همین طریق توانستیم اسممان را با کدینگ سزار با کلید M کد کنیم.

Mahyar Onsori -> Ymtkmd Azeadu (with key='M')

2. Encipher the following quote using the substitution cipher, use the given cipher alphabet as the key and the remainder of your student number divided by 26 as the offset. (to do this exercise select Crypt/Decrypt > Symmetric (classic) > Substitution/Atbash).

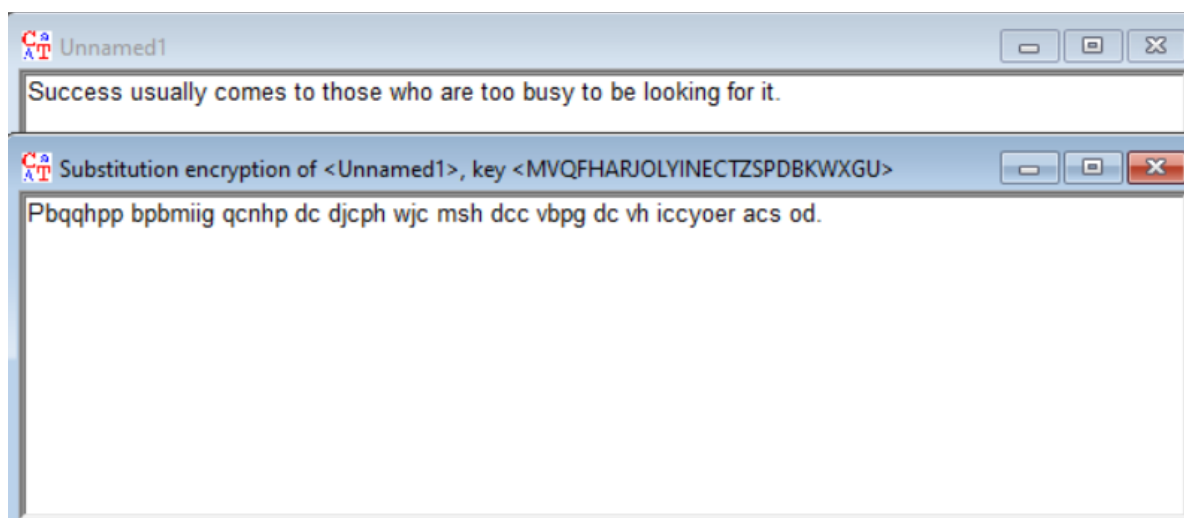
Plain text: Success usually comes to those who are too busy to be looking for it.

Cipher alphabet: fhajolyinctzspdbkwxgumvq



The dialog box is titled "Key Entry: Monoalphabetic Substitution / Atbash". It contains three sections: "Choose variant of the monoalphabetic substitution" with three radio buttons (the first is selected), "Key Input" with a "Key:" field containing "FHARJOLYINCTZSPDBKWXGUMVQ" and an "Offset:" field containing "3", and "Information on the substitution encryption" showing a mapping from "ABCDEFGHIJKLMNOPQRSTUVWXYZ" to "MVQFHARJOLYINCTZSPDBKWXGU". At the bottom are buttons for "Encrypt", "Decrypt", "Text options", and "Cancel".

$$9632093 \bmod 26 = 3$$



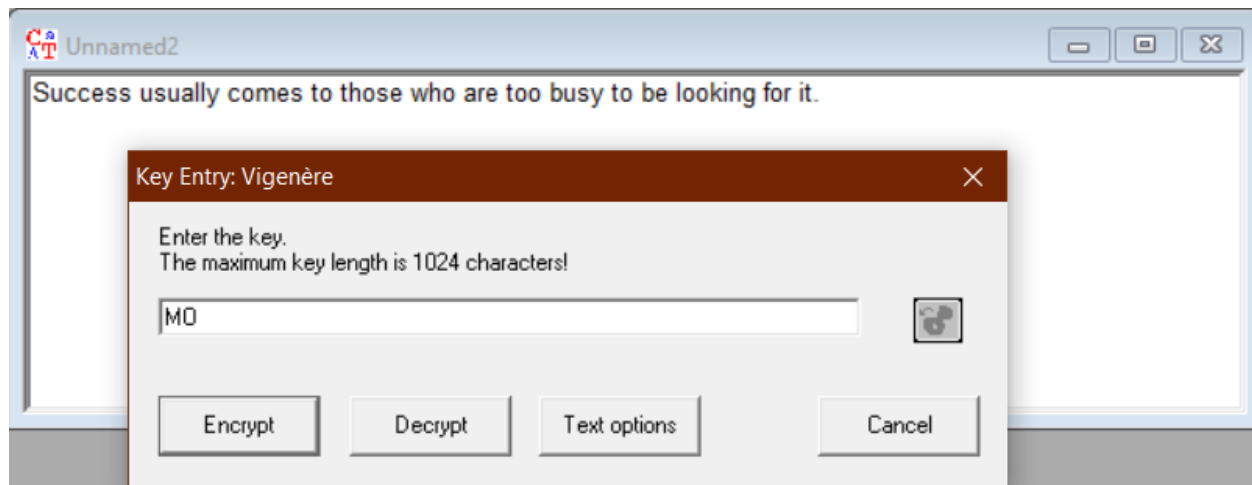
همانگونه که در صورت سوال گفته شده است از روش جانشینی استفاده می شود.

3. “Vigenere” Cipher is a method of encrypting alphabetic text. by using a series of interwoven Caesar ciphers, based on the letters of a keyword. It uses a simple form of polyalphabetic substitution. A polyalphabetic cipher is any cipher based on substitution. The encryption of the original text is done using the Vigenère square or Vigenère table.

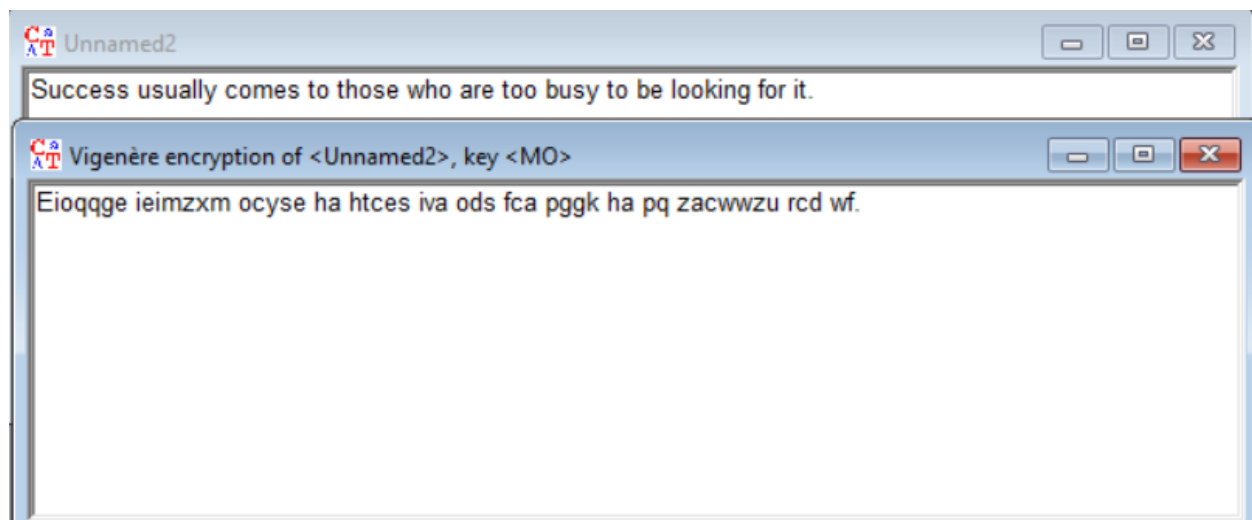
First described by Giovan Battista Bellaso in 1553, the scheme was misattributed to Blaise de Vigenère (1523–1596) in the 19th century, and so acquired its present name.

(To encipher your text using this method select Crypt/Decrypt > Symmetric (classic) > Vigenère)

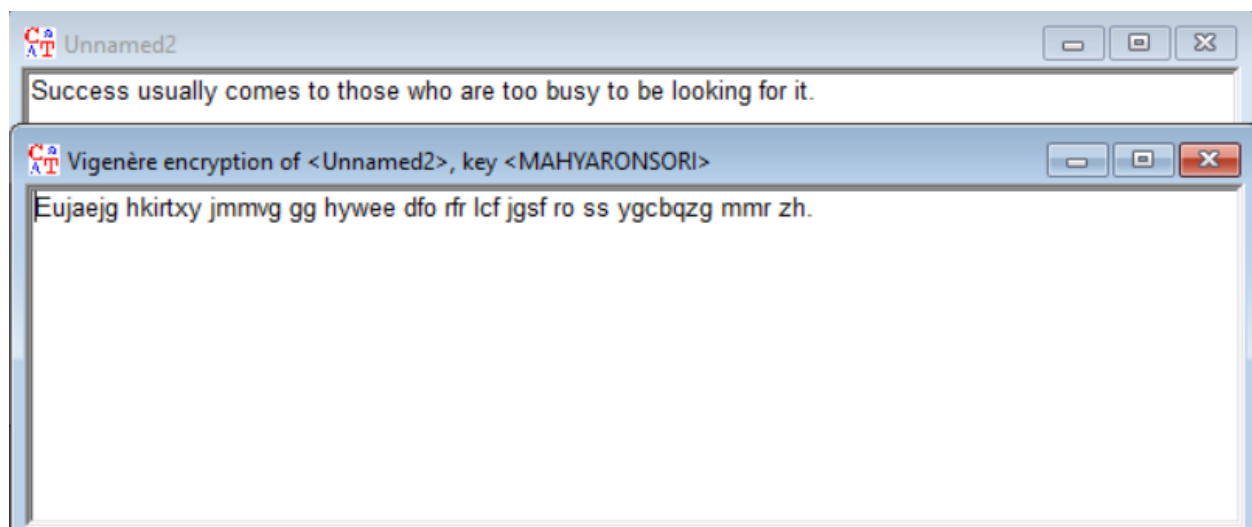
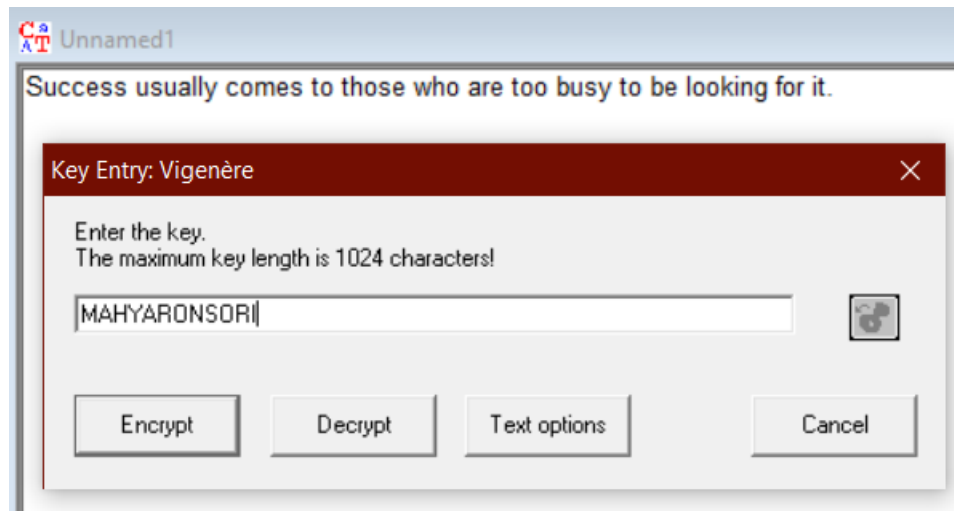
- a. Derive a bilateral key by concatenating the first letters of your first name and family name and encrypt the plain text used in the previous question using the Vigenere Cipher with this key.



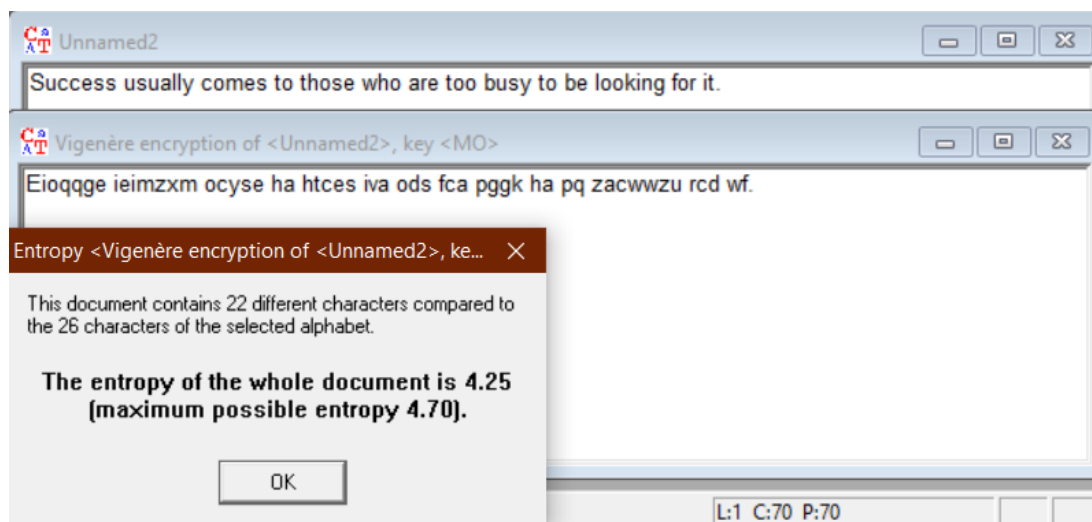
Mahyar Onsori



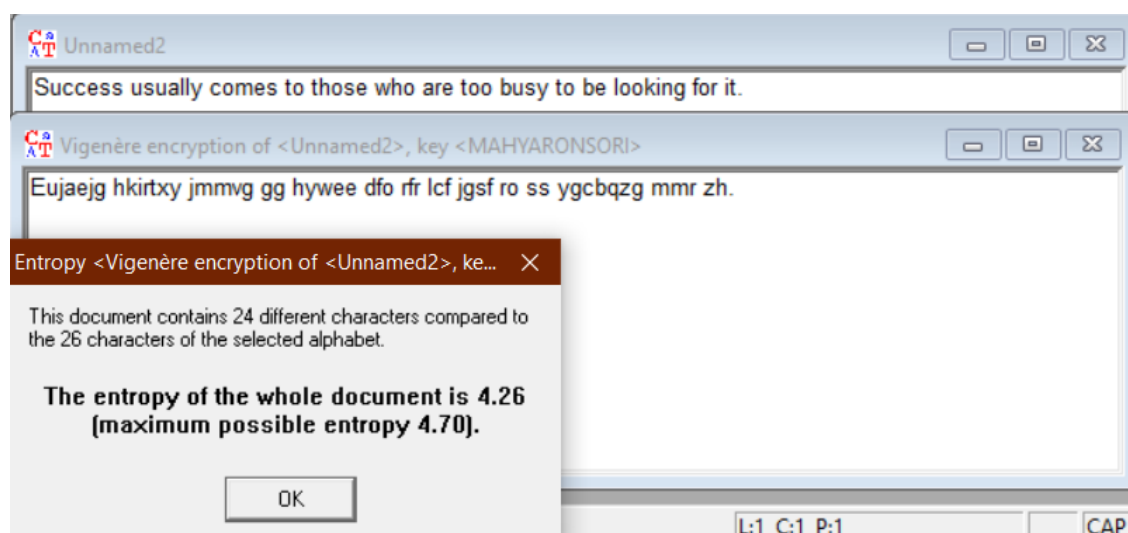
- b. Encrypt the same text using the same algorithm, but this time, generate the key by concatenating your full first name with your last name.



- c. Compare the results of previous parts by analyzing their entropy. What do you think the entropy is? According to this measure, how does the key length affect the cipher text? Explain your reasons. (to do this exercise you can use Analysis > Tools for Analysis > Entropy)



With key = 'MO'



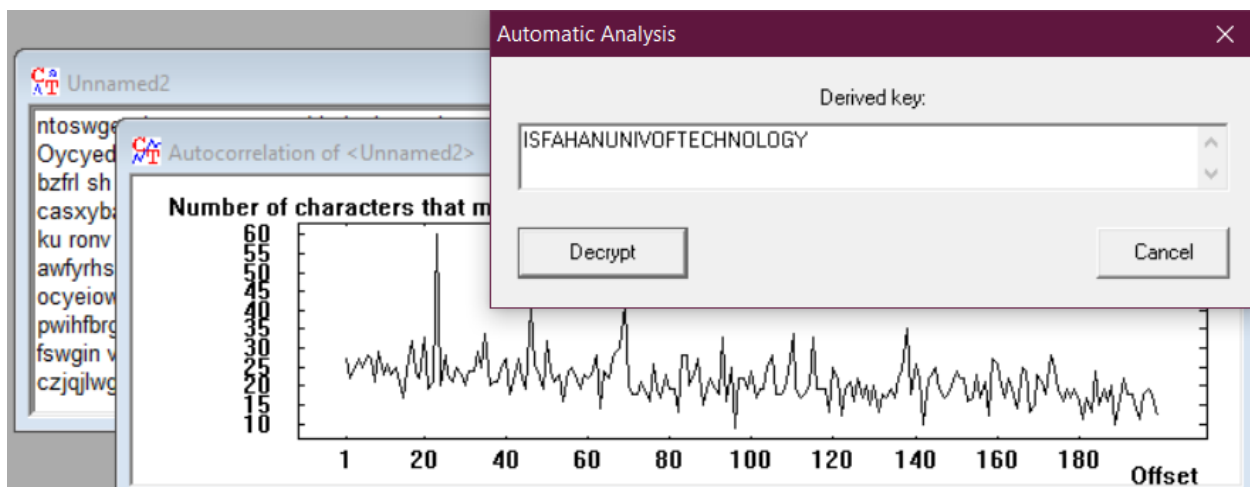
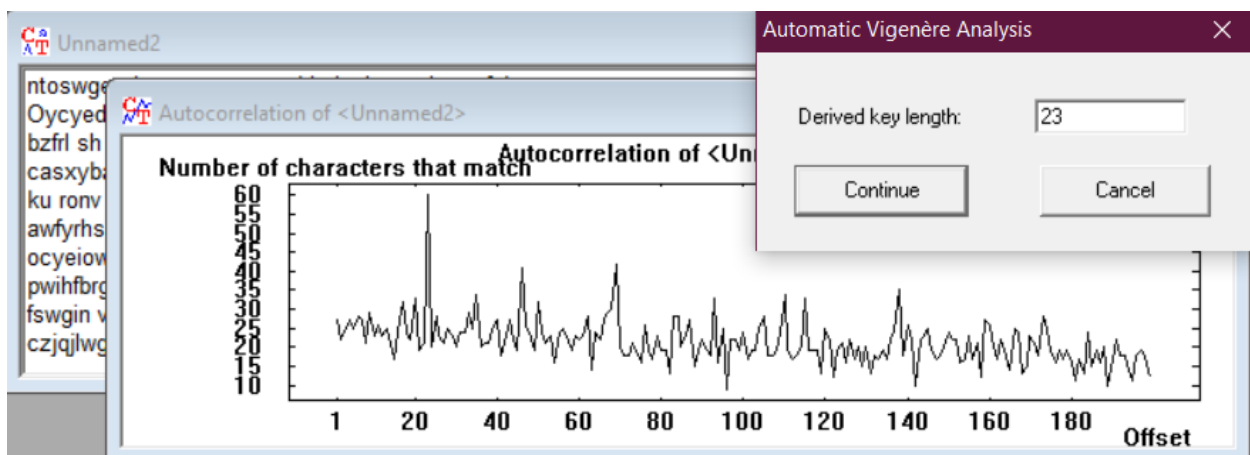
With key = 'MAHYARONSORI'

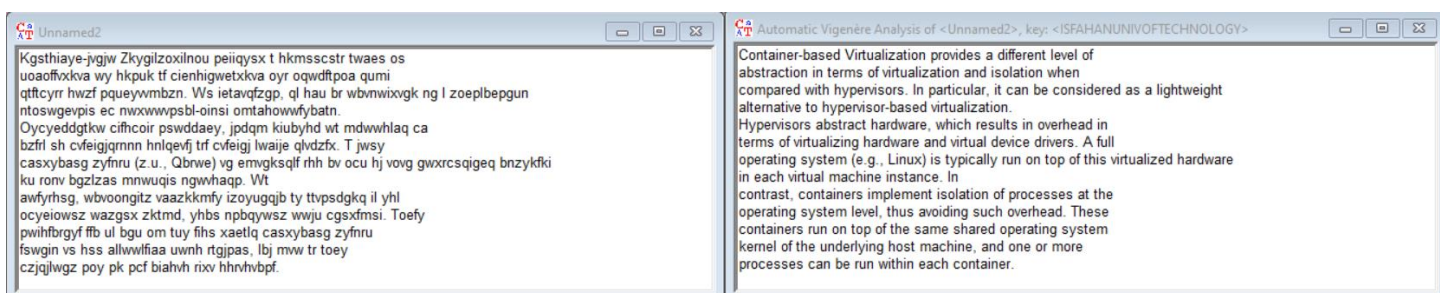
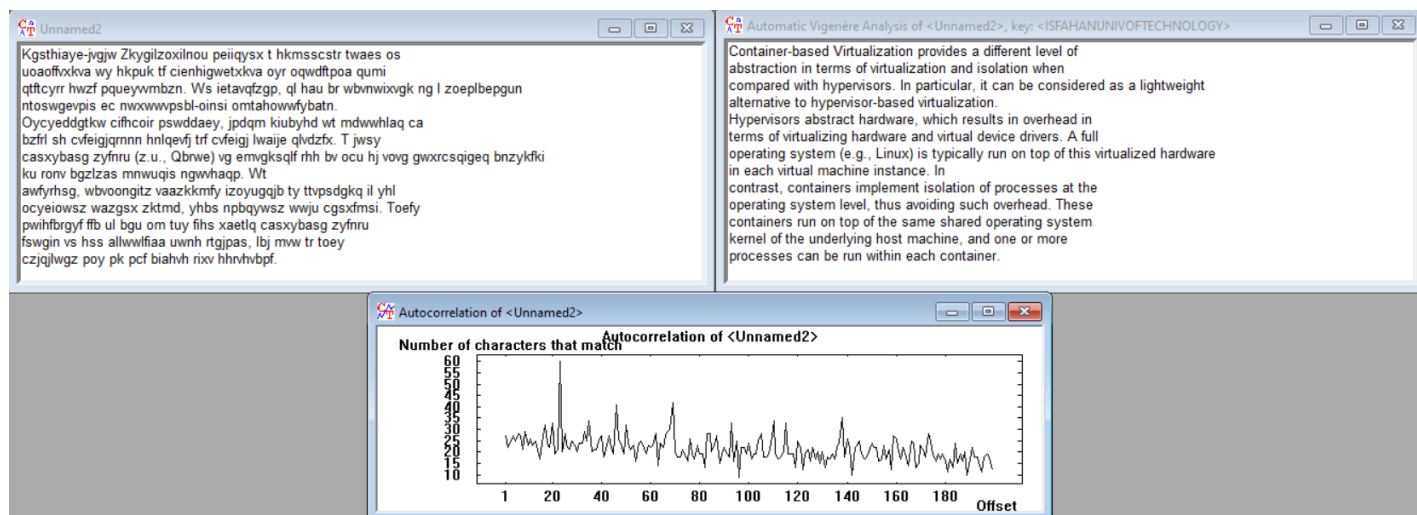
با زیاد شدن طول کلید آنتروپی بیشتر شد. زیاد شدن آنتروپی موجب زیاد شدن اطلاعات پیام می شود. از طرفی چون با افزایش طول کلید رمزگذاری بهتر انجام می شود پس احتمال رمزگشایی کم می شود و اطلاعات پیام بیشتر می شود (رابطه احتمال و اطلاعات عکس و رابطه اطلاعات و آنتروپی مستقیم است) و آنتروپی زیاد می شود.

4. Decipher the following cipher text, enciphered with Vigenere cipher, using CrypTool analytical tools, what do you guess the drawn diagram is?

(To break the cipher select Analysis > symmetric Encryption (classic) > Ciphertext-only > Vigenere)

Kgsthiaye-jvgjw Zkygilzoxilnou peiiqysx t hkmsscstr twaes os
uoaoffvxkva wy hkpuk tf cienhigwetxkva oyr oqwdftpoa qumi
qtftcyrr hwzf pqueyvmbzn. Ws ietavqfzgp, ql hau br wbnwixvgk ng l zoepbepgun
ntoswgevpis ec nwxwwvpsbl-oinsi omtahowwfybatn.
Oycyeddgkw cifhcoir pswddaey, jpdqm kiub yhd wt mdwwhlaq ca
bzfrl sh cvfeigjqrnnn hnlqevfj trf cvfeigj lwaije qlvdzfx. T jwsy
casxybasg zyfnru (z.u., Qbrwe) vg emvgksqlf rhh bv ocu hj vovg gwxrcsqigeq bnzykfki
ku ronv bgzlas mnwugis ngwvhaqp. Wt
awfyrhsg, wbvoongitz vaazkkmfy izoyugqjb ty ttvpsdgkq il yhl
ocyeiosz wazgsx zktmd, yhbs npbqywsz wwju cgsxfmsi. Toefy
pwihfbrgyf ffb ul bgu om tuy fihs xaetlq casxybasg zyfnru
fswgin vs hss allwwlfiaa uwnh rtgipas, lbj mvw tr toey
czjqilwgz poy pk pcf biahvh rixv hhrvhvbpf.





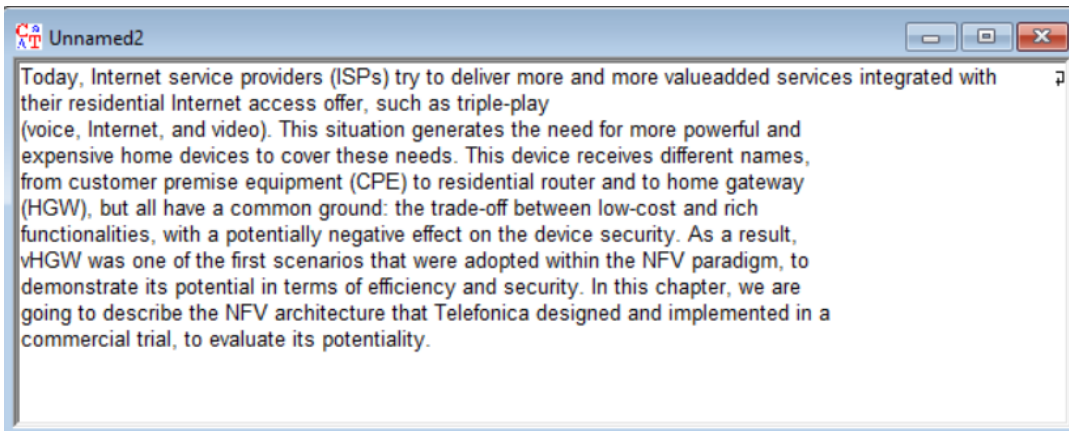
به نظر در جایی که نمودار AutoCorrelation پیک
میزند بهترین طول برای کلید را پیدا کرده است زیرا
بیشترین میزان همبستگی را دارد و بنابراین بهترین
طول کلید که سیستم حدس میزند در آنجا خواهد بود.

5. In cryptography, the one-time pad is an encryption technique that cannot be cracked, but requires the use of a one-time pre-shared key the same size as, or longer than, the message being sent. In this technique, a plaintext is paired with a random secret key. Answer the following questions regarding this encryption technique.

- With the help of Cryptool, encrypt the following plaintext using the given key as one time pad. (to do this exercise select Crypt/Decrypt > Symmetric (classic) > OTP)

Plaintext: Today, Internet service providers (ISPs) try to deliver more and more value-added services integrated with their residential Internet access offer, such as triple-play (voice, Internet, and video). This situation generates the need for more powerful and expensive home devices to cover these needs. This device receives different names, from customer premise equipment (CPE) to residential router and to home gateway (HGW), but all have a common ground: the trade-off between low-cost and rich functionalities, with a potentially negative effect on the device security. As a result, vHGW was one of the first scenarios that were adopted within the NFV paradigm, to demonstrate its potential in terms of efficiency and security. In this chapter, we are going to describe the NFV architecture that Telefonica designed and implemented in a commercial trial, to evaluate its potentiality.

OTP Key: Thksp, Afuqfgwj abnqkku xdhsqlbfo (TYTd) jom th kifbxdy sevo kxr atlj wedxe-utpxk yehxmtsb gfuqyzfbjo smpx trwul vmnqlbjqcsp Nfuqfgwl qxarge otlrgh, evsf ed teqjoi-fswi (zhwte, Kuqqeyeb, fbb lqlbm). Iwsh yoxbwdpif lstjhrccq cndu dpq sevg thouvznz oes igmgfzmxz tcva vnxmtqq ne hqxdy xkiui vgohc. Ykqw fadxms dgdqmxzi vjfwedryj seofc, vuyc towrfagz zsqzqzq bugijwnkz (NES) nv vmnqlbjqcog osodqr lzd fa tber sshdgtc (ZEX), fbq imw vbmq v quhgif lpipzd: fvx urhga-kjd cqlupmm ecp-yeey oes hwfd oufliysvsnrcfy, zeto w tbsuecccfpp iebsvlei mnjgbd xo xki hfinwte wtoknvni. Xw k tepqvb, vZEX azk ghf ee whl eypbh cvoxxtkpc ykqp funx uvdnckd exbguv rzg SMF ksrhggfq, th kirqnmmhrcy sru xrlzsngh ne pveeb az dklapguzsf wle cwnmlfey. Ar bgua ejysimv, zs gis nikhd th kmhotxcy uag SMV jkydtnklxxy uaus Wejptzesxs riduetwz bbo ygcaymbfuqb hc a dgehoxmpqe ufyzh, ca slznihte dhp iwljnqcsphj.U

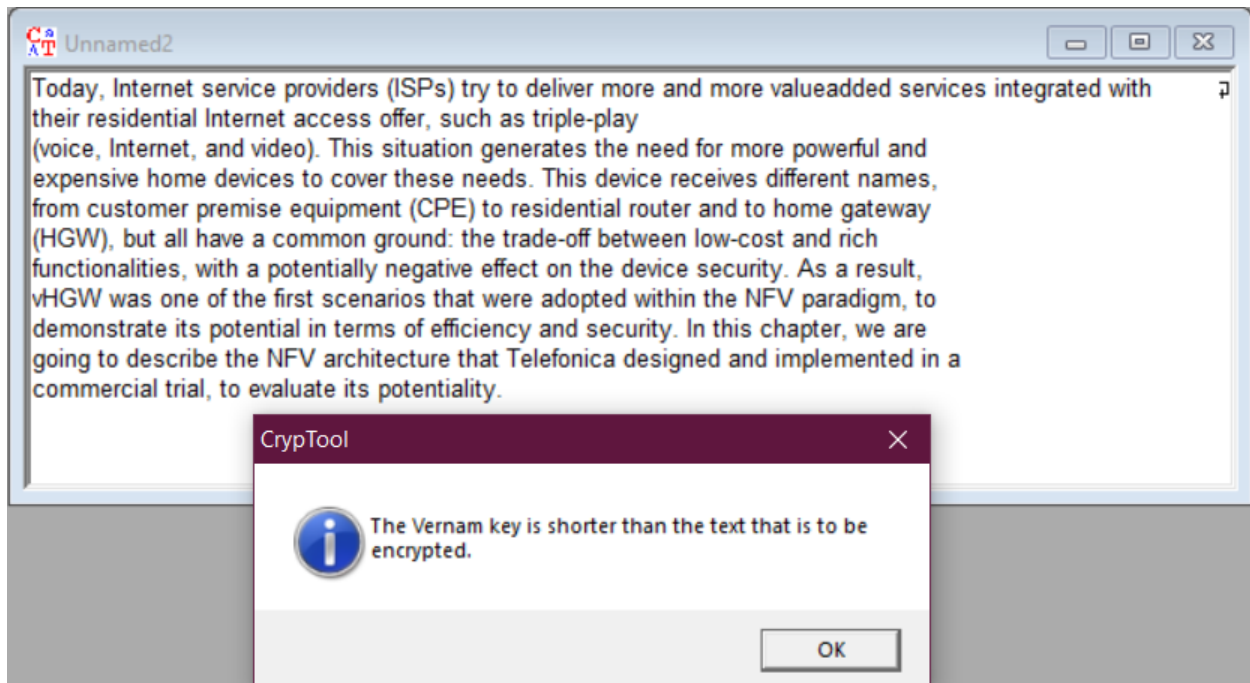


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0000002C 00 00 07 00 0F 0C 0A 0E 01 0B 00 1E 0A 04 0A 00 0A 16 16 00 0C 1B 1E 0F 2D 7C 16 09 11 1D 04 49 11 11 14 58 18 45 0B 13 01 1B 08
07 53 0B 4E 13 03 12 03 18 0E 03 06 4A 18 49 07 05 00 0C 48 11 1B 05 5E 1E 45 05 04 0A 14 02 16 03 10 0F 53 39 4E 3A 03 07 1F 03 13
00000058 57 0D 43 12 1D 12 01 47 0A 46 09 11 1E 5E 55 5F 78 69 0D 56 12 15 00 11 16 49 04 09 14 47 1F 05 4C 1F 7E 7D 41 56 47 13 0B 12 58 45
00000084 65 4E 3F 10 03 1F 00 0D 49 42 4D 4E 02 42 14 49 08 14 03 4B 43 09 7A 48 20 04 53 1B 49 0D 1A 19 16 1E 0B 1E 49 01 45 02 16 06 0B 1C
000000B0 17 10 51 05 48 0B 50 0A 45 06 03 4F 0E 62 78 44 1D 1E 52 16 45 06 1E 57 11 1A 09 00 1A 4E 1B 09 44 62 6F 16 58 19 02 03 14 0F 0C 08
000000DC 58 12 4F 19 06 56 05 45 00 07 1B 0B 07 51 05 4F 4E 06 4F 1E 1A 0A 44 0D 48 1D 18 0C 55 07 45 13 03 1C 46 43 7A 48 30 1B 51 13 45 10
00000108 08 07 1D 4D 01 68 69 01 0E 12 14 1E 58 1E 00 46 10 0F 14 12 0B 10 52 17 0B 4D 16 16 43 6B 69 4A 52 19 18 59 00 55 07 1B 18 1F 03 13
00000134 47 0A 52 1F 1E 18 02 1F 51 02 51 17 1C 17 04 0F 19 1F 4E 52 63 78 0B 6C 73 5D 4F 4E 04 45 05 04 0A 14 02 16 03 10 0F 4F 15 62 7F 1B
00000160 16 1D 44 10 1C 44 4C 0E 0B 00 0E 0E 4D 11 42 04 13 54 16 04 09 1D 6A 7E 4B 68 6F 0D 6C 74 09 4E 55 12 42 10 4C 05 4D 1F 41 00 07 4D
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000001B8 1F 5E 0D 45 04 04 44 4F 17 1A 43 00 7A 6C 02 55 01 16 12 05 06 17 12 1A 1A 04 07 17 10 4A 59 5B 49 0E 0D 54 0E 00 07 4F 00 07 1D 01
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00000220 40 56 28 5D 00 39 57 52 0E 53 01 09 04 5D 6F 7C 5A 68 31 0D 65 2F 41 12 5A 04 4E 02 48 09 46 45 11 48 12 48 0A 49 17 0A 04 42 1B 43
0000023C 06 18 0E 0A 11 1B 18 50 17 48 1B 1F 4B 07 45 14 10 4E 19 69 65 05 02 01 0A 43 1C 0D 15 4D 11 0C 47 01 1E 45 52 34 21 76 73 3D 27 52
00000268 0A 17 1B 0F 0A 4B 46 05 43 2D 7E 0C 45 06 06 1C 02 1A 1C 0C 1C 17 43 10 54 00 52 05 4F 0C 17 02 0E 1A 0F 0B 54 01 4E 4E 11 45 02 1B
00000294 16 45 0D 46 41 1F 6B 6C 0D 08 05 04 1E 04 0C 5A 12 08 44 57 1F 00 43 16 05 07 19 15 48 45 3C 40 00 35 1A 49 11 47 16 09 41 15 1E 16
000002C0 0B 45 4D 01 49 00 1B 01 45 6A 60 14 4F 07 07 0C 48 10 4F 54 0C 45 18 0E 1A 06 16 1D 43 0D 48 10 41 29 66 05 6D 37 7F 69 02 02 0D 01
000002EC 17 1A 1E 1E 1D 58 1D 11 41 01 41 21 16 4C 32 03 05 1E 13 17 04 53 1C 16 53 1B 0E 0A 10 01 54 16 14 44 42 0B 02 50 15 02 0E 04 17 16
00000318 08 02 55 18 0C 00 09 6E 2A 02 4F 09 0A 00 1A 0C 11 0C 1C 51 11 52 1C 07 15 56 48 58 4F 43 04 7B 6B 1F 19 1B 1A 0C 48 1D 11 53 44 18
00000344 1F 54 0C 19 18 07 0B 1D 0A 07 09 40
00000370

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- b. Use your full name as the OTP key and encrypt the same plaintext with the same algorithm. (don't forget to write the key in your answer file)

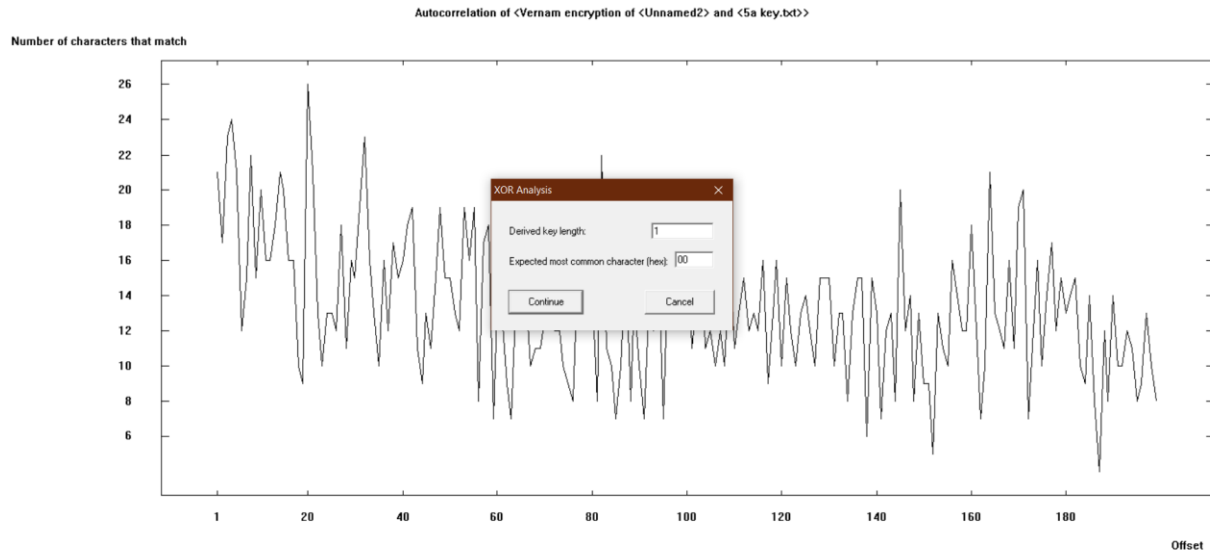


Key = 'MAHYARONSORI'

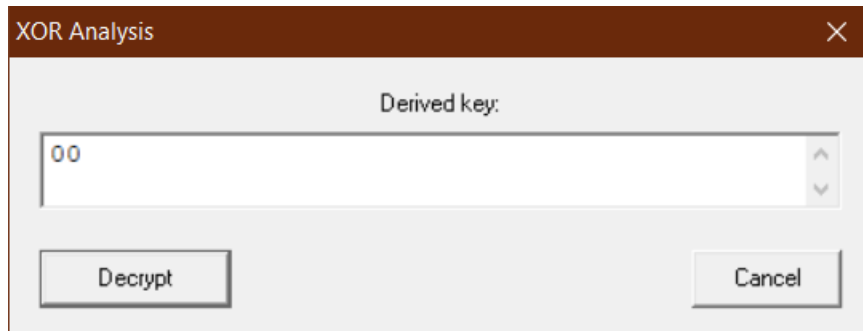
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CrypTool 1.4.41 - [Vernam encryption of <Unnamed2> and <5b key.txt>]
File Edit View Encrypt/Decrypt Digital Signatures/PKI Indiv. Procedures Analysis Options Window Help

00000000 19 2E 2C 38 38 7E 6F 07 3D 3B 37 3B 23 24 3C 79 32 37 3D 38 3A 2C 37 69 3D 33 27 2F 28 36 2A 3C 20 6F 7A 00 1E 11 3B 70 61 26 3D 37 ...88"o .:7:#$<y27=8..7i=3'/(6* < cz...pa&=7
00000002 73 3B 3D 69 29 24 24 30 37 37 3D 6E 3E 20 20 2C 6D 20 26 3D 61 3F 20 3C 36 6F 24 28 21 34 2D 38 25 36 2A 2A 73 3C 37 3B 3B 28 2B 3C s:=i)$8077"n> .n &a=? <6o8(14-8%6**s<7:.(+<
00000008 32 72 26 20 27 2A 35 3B 2C 35 2D 3D 61 25 26 3A 3B 6F 26 21 28 28 3A 79 33 37 3C 27 37 2A 3C 3D 24 20 24 79 08 3C 3B 2B 21 21 37 3D 2rk &5. S--a&%.o&l((.y37:'7*;<s $y .:i|7-
0000000A 6D 20 2B 3A 24 21 3C 6E 3C 29 34 2C 3F 6D 68 2A 34 31 27 6E 32 3C 72 3D 3F 28 38 35 24 7F 3F 22 32 36 5F 43 65 37 27 30 22 37 63 6E m +=s!(<x)4.7mh*41n2<r=?($5$ 7*26.Ce/0*7cm
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00000010 3A 2C 37 69 3F 24 2B 3C 28 24 2A 3D 73 2B 3B 2F 2B 24 3A 3C 2F 26 6F 20 32 22 37 3A 61 4C 42 3F 33 3D 22 6E 3D 3A 21 3D 22 2C 2B :7i78+<($**+;/e$< <6o 2*7.aLB78**nd1=+.
00000012 61 22 3D 2B 3E 26 21 2C 6D 24 39 2C 28 22 22 2B 3D 3B 72 61 0E 11 0D 70 61 26 20 6E 21 2A 21 20 29 24 26 2D 28 33 23 6E 21 20 27 3D a"+&l n$9.('"+&:a. pa& n1*)$&-(3#n1 '-
00000014 28 33 68 3B 2F 36 6F 3A 3C 6F 3A 26 20 24 68 3E 20 26 2A 39 32 36 5F 43 65 09 0F 0E 68 7E 6F 2C 26 3B 72 28 21 2D 68 31 20 24 2A 6E (3h8/6o.co/& $h) &*926.Ce...h'o.&r(l-h1 $*n
00000016 32 6F 31 26 20 2C 27 37 61 35 3D 21 26 21 36 73 6D 35 20 3C 61 26 3D 2F 37 2A 7F 26 2B 27 68 3B 24 26 38 2B 36 21 72 25 22 36 65 3A 2o1& .7a5=1&l6sm5 <a&+7* &+h;8&8+61z'6e:
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0000001A 24 20 24 35 38 72 21 2B 34 2E 26 20 3B 24 68 3C 27 34 2A 2D 27 6F 3D 27 6D 35 20 3C 61 36 2A 38 3A 2C 37 69 3E 24 2B 2C 33 3B 3B 37 $ $58x144 & .9h<'4*-o'm5 <a&8:7i>6+3:;7
0000001C 7D 6F 13 3A 6D 20 68 2B 24 21 3A 22 27 63 5F 43 3B 09 0F 0E 61 25 2E 3D 73 20 3C 2C 6D 2E 2E 79 35 3A 2A 6E 35 26 20 3A 39 61 3B 3A }o.m h+8l:'c.C.:.a% <s <.m. y5.*n5&:9a:
0000001E 24 3C 2E 3C 3A 20 21 69 39 29 29 2D 61 25 2A 3C 36 6F 33 2D 22 31 3C 3C 25 72 38 27 27 3B 27 6D 35 20 3C 61 1C 09 18 73 3F 33 3B $<(<: l19))=a%*6o3="1<<xr8'''. n5 <a. s73:
0000001F 2C 25 21 3E 2C 7E 6F 3A 3C 42 58 2D 28 2C 27 37 32 26 3D 2F 27 2A 72 20 39 32 68 29 2E 26 2A 20 27 26 33 25 6D 28 26 79 35 37 3D 23 .4i>.co.<BX(-.72&+/* 92h) &* 'k35a(&y57*#
00000020 20 6F 3D 2F 6D 24 2E 3F 28 31 26 2B 3C 2C 2B 69 2C 2F 2C 79 32 37 2C 3B 21 26 26 30 63 61 01 37 61 26 27 27 20 6F 31 21 2C 31 3C 3C o=/m8 ?(1&+.i./y27.:l&&0ca.7a&' o1l.1<<
00000022 33 7E 6F 39 36 6F 33 3B 28 4C 42 3E 2C 3B 21 29 73 3B 3D 69 29 24 3B 3A 33 3B 2D 2B 73 3B 3A 2C 6D 0F 0E 0F 61 33 3D 2D 3B 26 26 2C 3'o96o3.(LB):|)s:=i)$:;+s:;...a3=-;&&.
00000024 2E 35 3D 2B 24 72 3B 26 32 3B 72 1D 28 2D 2D 3F 2E 3C 26 2D 32 6F 36 2C 3E 28 2F 37 24 36 6F 2F 3D 2B 72 20 20 31 24 3C 2C 37 21 3A 5+=&r;&2.r.(--?<k-2o6)/786o/+r 18<7l:
00000026 36 2B 72 20 23 61 29 54 4B 31 20 23 3E 2A 20 2A 24 20 24 79 35 20 26 2F 3F 63 72 3D 22 61 2D 2F 20 3E 3A 2F 27 2A 72 20 39 32 68 29 6+r #a)TKI #>* $ $y5 &/7cr="a-/./*r 92h)
00000028 2E 26 2A 20 27 26 33 25 24 35 31 77 .&* &3%$51v
```

- c. Try to find the OTP key of each of the two cipher texts in the previous parts using the Cryptool analytical tools and put the results, especially the predicted length of the key, in your answer file; Compare the security of the keys based on the results of analysis. (To do this part select Analysis > Symmetric Encryption > cipher text only > XOR)



Predicted Key Length = 1



Cryptool 1.4.41 - [Automatic XOR Analysis of <Vernam encryption of <Unnamed2> and <5a key.txt> -> key: <00>]

File Edit View Encrypt/Decrypt Digital Signatures/PKI Index Procedures Analysis Options Window Help

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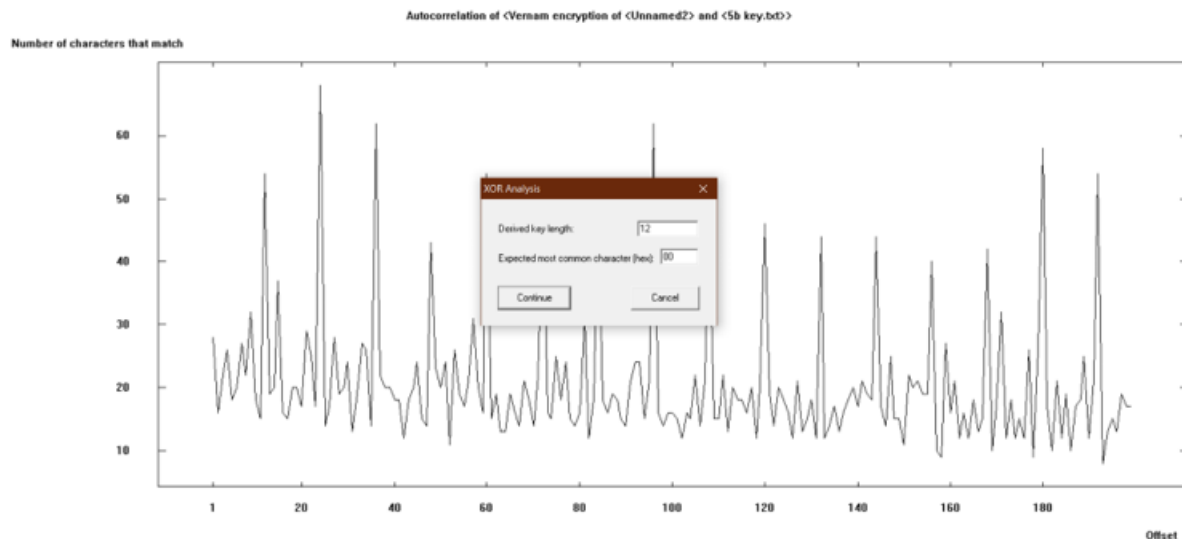
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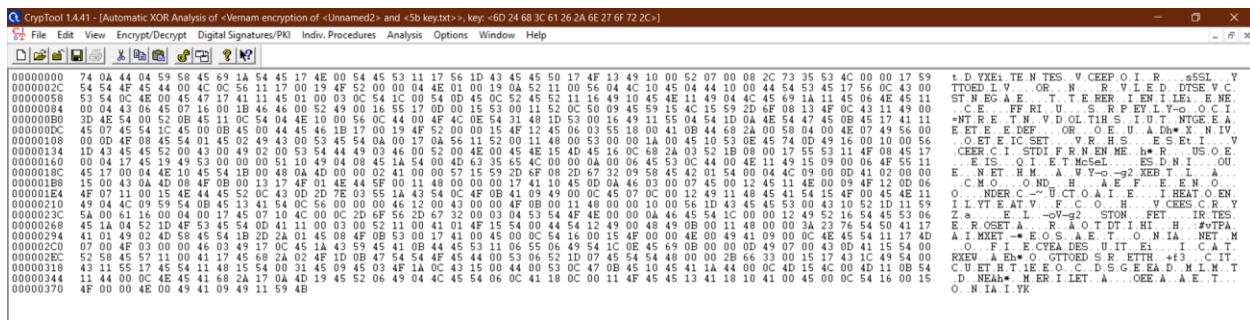
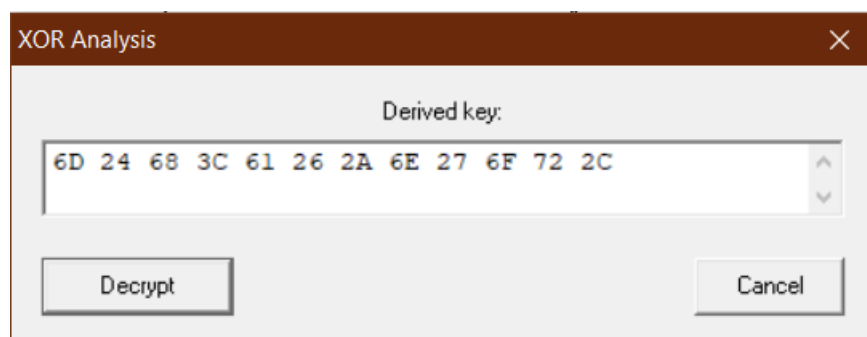
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e.N.....I.B.N.B.I..K.C.H..S.I.....I..E
X.Q.H.P.E..O.b.x.D..R.E.W.....N..D.b.o.I
X.Q.V.E.....Q.O.N.O...D.H..U.E..F.C.Z.H.O.Q.E
M.hi.....X.F.....R..M..C.h.i.R..Y.U
G.R.....O.Q.....N.R.c.x..l.s.J.O.N.E.....O.b
D..D.L..H.B.T.....K.h.o..l.t.N.U.B.I.M.A.M
O.....I.R.....Q.D.H.V.R..J.K.J.h.....P.U.H
A.E.D.O..C.z.l.U.....J.V.I.T..O
P.E.....E.F..B.N.I.e.o.X.v..U.C
@V(I).S.V.S..J.o.I.z.h.l.e.A.Z.W.H.F.E..H.H.I..B.C
P.H.K.E.N.a.e..C.T..G.E.R.4.i.v.s'R
K.F.C~E.....C.T.R.O.....T.N.N.E
E.F.A.kl.....Z..D.V.C..H.E@S.I.G.A
E.H.I.Ej.O..H.O.T.E.....C.H.A)i.a.7.i
X.A.A'i.L2.....S.S..T.D.B..P
T.....a*.O.....Q.R..V.H.X.O.C.(k.....H..S.D
T.....@

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Predicted Key Length = 12



علت اینکه طول کلید را در حالت اول ۱ حدس زده است به این دلیل است که برنامه نتوانسته کلید را تشخیص دهد. در حالت دوم طول کلید را درست حدس میزند اما خود کلید را اشتباه تشخیص می دهد.